For the kernels $K_1, \dots, K_d$ satisfying the properties presented in Section 3.1, we define the constants

$$\kappa_1(d) \coloneqq \prod_{i=1}^{d} \int_{\mathbb{R}} |K_i(x_i)| \mathrm{d}x_i < \infty \quad \text{and} \quad \kappa_2(d) \coloneqq \prod_{i=1}^{d} \int_{\mathbb{R}} K_i(x_i)^2 \,\mathrm{d}x_i < \infty \tag{21}$$

which are well-defined as $K_i \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ for $i = 1, \dots, d$ by assumption. We do not make explicit the dependence on $K_1, \dots, K_d$ in the constants as we consider those to be chosen *a priori*. Moreover, we often use the kernel properties of k_λ presented in Equation (8), that are

$$\int_{\mathbb{R}^d} k_\lambda(x, y) \mathrm{d}x = \prod_{i=1}^{d} \frac{1}{\lambda_i} \int_{\mathbb{R}} K_i\left(\frac{x_i - y_i}{\lambda_i}\right) \mathrm{d}x_i = \prod_{i=1}^{d} \int_{\mathbb{R}} K_i(x_i') \mathrm{d}x_i' = 1$$

and

$$\int_{\mathbb{R}^d} k_\lambda(x, y)^2 \mathrm{d}x = \prod_{i=1}^{d} \frac{1}{\lambda_i^2} \int_{\mathbb{R}} K_i\left(\frac{x_i - y_i}{\lambda_i}\right)^2 \mathrm{d}x_i = \frac{1}{\lambda_1 \cdots \lambda_d} \prod_{i=1}^{d} \int_{\mathbb{R}} K_i(x_i')^2 \,\mathrm{d}x_i' = \frac{\kappa_2}{\lambda_1 \cdots \lambda_d}.$$

We often use in our proofs the standard result that, for $a_1, \dots, a_\ell \in \mathbb{R}$, we have

$$\left(\sum_{i=1}^{\ell} a_i\right)^2 \leq \left(\sum_{i=1}^{\ell} 1^2\right)\left(\sum_{i=1}^{\ell} a_i^2\right) = \ell \sum_{i=1}^{\ell} a_i^2$$

which holds by Cauchy–Schwarz inequality.

In our proofs, we show that there exist some constants which are large enough so that our results hold. We keep track of those constants and show how they depend on each other. The aim is to show that such constants exist, we do not focus on obtaining the tightest constants possible.

E.1 Proof of Proposition 1

Recall that in Sections 3.2.1 and 3.2.2 we have constructed elements $\big(\widehat{M}_\lambda^b\big)_{1 \leq b \leq B+1}$ for the two MMD estimators defined in Equations (3) and (6), respectively. The first one uses permutations while the second uses a wild bootstrap. For those two cases, we first show that the elements $\big(\widehat{M}_\lambda^b\big)_{1 \leq b \leq B+1}$ are exchangeable under the null hypothesis $\mathcal{H}_0 \colon p = q$. We are then able to prove that the test $\Delta_\alpha^{\lambda,B}$ has the prescribed level using the exchangeability of $\big(\widehat{M}_\lambda^b\big)_{1 \leq b \leq B+1}$.

Exchangeability using permutations as in Section 3.2.1.

Recall that in this case we have permutations $\sigma^{(1)}, \dots, \sigma^{(B)}$ of $\{1, \dots, m+n\}$. We also have $\widehat{M}_\lambda^b \coloneqq \widehat{\mathrm{MMD}}^2_{\lambda,\mathtt{a}}(\mathbb{X}_m^{\sigma^{(b)}}, \mathbb{Y}_n^{\sigma^{(b)}})$ for $b = 1, \dots, B$ and $\widehat{M}_\lambda^{B+1} \coloneqq \widehat{\mathrm{MMD}}^2_{\lambda,\mathtt{a}}(\mathbb{X}_m, \mathbb{Y}_n)$. Following the same reasoning as in the proof of Albert et al. (2022, Proposition 1, Equation C.1), we can deduce that $\big(\widehat{M}_\lambda^b\big)_{1 \leq b \leq B+1}$ are exchangeable under the null hypothesis. The only difference is that they work with the Hilbert Schmidt Independence Criterion rather than with the Maximum Mean Discrepancy, but this does not affect the reasoning of the proof.

Exchangeability using a wild bootstrap as in Section 3.2.2.

The exchangeability under the null using the wild bootstrap follows from the one using permutations since by Proposition 11 using the wild bootstrap corresponds to using a subgroup of the permutation group. We show below how the original statistic and the permuted one can be seen to have the same distribution under the null.

For $b = 1, \dots, B$, we have n i.i.d. Rademacher random variables $\epsilon^{(b)} \coloneqq \big(\epsilon_1^{(b)}, \dots, \epsilon_n^{(b)}\big)$ with values in $\{-1,1\}^n$ and

$$\widehat{M}_\lambda^b \coloneqq \frac{1}{n(n-1)} \sum_{1 \le i \ne j \le n} \epsilon_i^{(b)} \epsilon_j^{(b)} h_\lambda(X_i, X_j, Y_i, Y_j)$$

where h_λ is defined in Equation (5). We also have

$$\widehat{M}_\lambda^{B+1} \coloneqq \widehat{\mathrm{MMD}}^2_{\lambda,\mathtt{b}}(\mathbb{X}_n, \mathbb{Y}_n) = \frac{1}{n(n-1)} \sum_{1 \le i \ne j \le n} h_\lambda(X_i, X_j, Y_i, Y_j).$$

By the reproducing property of the kernel k_λ, we have

$$\begin{aligned}
\left(\sup_{f \in \mathcal{F}_\lambda} \frac{1}{n} \sum_{i=1}^n \big(f(X_i) - f(Y_i)\big) \right)^2 &= \left(\sup_{f \in \mathcal{F}_\lambda} \left\langle f, \frac{1}{n} \sum_{i=1}^n k_\lambda(X_i, \cdot) - \frac{1}{n} \sum_{i=1}^n k_\lambda(Y_i, \cdot) \right\rangle_{\mathcal{H}_{k_\lambda}} \right)^2 \\
&= \left\| \frac{1}{n} \sum_{i=1}^n k_\lambda(X_i, \cdot) - \frac{1}{n} \sum_{i=1}^n k_\lambda(Y_i, \cdot) \right\|^2_{\mathcal{H}_{k_\lambda}} \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n h_\lambda(X_i, X_j, Y_i, Y_j)
\end{aligned}$$

where $\mathcal{F}_\lambda \coloneqq \{f \in \mathcal{H}_{k_\lambda} : \|f\|_{\mathcal{H}_{k_\lambda}} \le 1\}$. Under the null hypothesis $\mathcal{H}_0 \colon p = q$, all the samples $(X_1, \dots, X_n, Y_1, \dots, Y_n)$ are independent and identically distributed. So, the distribution of $\big(\sup_{f \in \mathcal{F}_\lambda} \frac{1}{n} \sum_{i=1}^n \big(f(X_i) - f(Y_i)\big)\big)^2$ does not change if we randomly exchange X_i and Y_i for each $i = 1, \dots, n$. This can be formalized using n i.i.d. Rademacher random variables $\epsilon_1, \dots, \epsilon_n$, we have

$$\left(\sup_{f \in \mathcal{F}_\lambda} \frac{1}{n} \sum_{i=1}^n \big(f(X_i) - f(Y_i)\big) \right)^2 \overset{d}{\underset{\mathcal{H}_0}{=}} \left(\sup_{f \in \mathcal{F}_\lambda} \frac{1}{n} \sum_{i=1}^n \epsilon_i \big(f(X_i) - f(Y_i)\big) \right)^2,$$

where the notation $\overset{d}{\underset{\mathcal{H}_0}{=}}$ means that the two random variables have the same distribution under the null hypothesis $\mathcal{H}_0 \colon p = q$. Since we also have

$$\begin{aligned}
\left(\sup_{f \in \mathcal{F}_\lambda} \frac{1}{n} \sum_{i=1}^n \epsilon_i \big(f(X_i) - f(Y_i)\big) \right)^2 &= \left(\sup_{f \in \mathcal{F}_\lambda} \left\langle f, \frac{1}{n} \sum_{i=1}^n \epsilon_i k_\lambda(X_i, \cdot) - \frac{1}{n} \sum_{i=1}^n \epsilon_i k_\lambda(Y_i, \cdot) \right\rangle_{\mathcal{H}_{k_\lambda}} \right)^2 \\
&= \left\| \frac{1}{n} \sum_{i=1}^n \epsilon_i k_\lambda(X_i, \cdot) - \frac{1}{n} \sum_{i=1}^n \epsilon_i k_\lambda(Y_i, \cdot) \right\|^2_{\mathcal{H}_{k_\lambda}} \\
&= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \epsilon_i \epsilon_j h_\lambda(X_i, X_j, Y_i, Y_j),
\end{aligned}$$

we obtain

$$\sum_{i=1}^{n}\sum_{j=1}^{n} h_\lambda(X_i, X_j, Y_i, Y_j) \overset{d}{\underset{\mathcal{H}_0}{=}} \sum_{i=1}^{n}\sum_{j=1}^{n} \epsilon_i\epsilon_j h_\lambda(X_i, X_j, Y_i, Y_j).$$

Since $\epsilon_i^2 = 1$ for $i = 1, \dots, n$, subtracting $\sum_{i=1}^n h_\lambda(X_i, X_i, Y_i, Y_i)$ from both sides, we get

$$\sum_{1\leq i\neq j\leq n} h_\lambda(X_i, X_j, Y_i, Y_j) \overset{d}{\underset{\mathcal{H}_0}{=}} \sum_{1\leq i\neq j\leq n} \epsilon_i\epsilon_j h_\lambda(X_i, X_j, Y_i, Y_j).$$

Level of the test.

Following a similar reasoning to the one presented by Albert et al. (2022, Proposition 1), we have

$$\begin{aligned}
\Delta_\alpha^{\lambda,B}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_B) = 1 \quad &\iff \quad \widehat{\mathrm{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) > \widehat{q}_{1-\alpha}^{\lambda,B}\big(\mathbb{Z}_B \big| \mathbb{X}_m, \mathbb{Y}_n\big) \\
&\iff \quad \widehat{M}_\lambda^{B+1} > \widehat{M}_\lambda^{\bullet\lceil (B+1)(1-\alpha)\rceil} \\
&\iff \quad \sum_{b=1}^{B+1} \mathbb{1}\Big(\widehat{M}_\lambda^{b} < \widehat{M}_\lambda^{B+1}\Big) \geq \lceil (B+1)(1-\alpha)\rceil \\
&\iff \quad B+1-\sum_{b=1}^{B+1} \mathbb{1}\Big(\widehat{M}_\lambda^{b} < \widehat{M}_\lambda^{B+1}\Big) \leq B+1-\lceil (B+1)(1-\alpha)\rceil \\
&\iff \quad \sum_{b=1}^{B+1} \mathbb{1}\Big(\widehat{M}_\lambda^{b} \geq \widehat{M}_\lambda^{B+1}\Big) \leq \lfloor \alpha(B+1)\rfloor \\
&\iff \quad \sum_{b=1}^{B+1} \mathbb{1}\Big(\widehat{M}_\lambda^{b} \geq \widehat{M}_\lambda^{B+1}\Big) \leq \alpha(B+1) \\
&\iff \quad \frac{1}{B+1}\left(1+\sum_{b=1}^{B} \mathbb{1}\Big(\widehat{M}_\lambda^{b} \geq \widehat{M}_\lambda^{B+1}\Big)\right) \leq \alpha
\end{aligned}$$

where we have used the fact that $B+1-\lceil (B+1)(1-\alpha)\rceil = \lfloor \alpha(B+1)\rfloor$. Using the exchangeability of $\big(\widehat{M}_\lambda^{b}\big)_{1\leq b\leq B+1}$, the result of Romano and Wolf (2005a, Lemma 1) guarantees that

$$\mathbb{P}_{p\times p\times r}\left(\frac{1}{B+1}\left(1+\sum_{b=1}^{B} \mathbb{1}\Big(\widehat{M}_\lambda^{b} \geq \widehat{M}_\lambda^{B+1}\Big)\right) \leq \alpha\right) \leq \alpha.$$

We deduce that

$$\mathbb{P}_{p\times p\times r}\Big(\Delta_\alpha^{\lambda,B}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_B) = 1\Big) \leq \alpha.$$

E.2 Proof of Lemma 2

Let

$$\mathcal{A} \coloneqq \Big\{\widehat{\mathrm{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \leq \widehat{q}_{1-\alpha}^{\lambda,B}\big(\mathbb{Z}_B \big| \mathbb{X}_m, \mathbb{Y}_n\big)\Big\}$$